

COUNTEREXAMPLES TO THE ITERATED BEATTY RANGE CRITERION BELOW $5/4$, AND A SUFFICIENT PER-LEVEL CUTOFF

ABSTRACT. For irrational $\alpha > 1$ let $f_\alpha(y) = \lfloor \alpha y \rfloor$ on the positive integers. Khani, Valizadeh and Zarei prove that for irrational $\alpha > (3 + \sqrt{5})/2 \approx 2.618$ a positive integer x lies in the range of f_α^n if and only if an explicit system $\text{RC}_{\leq n}(\alpha, x)$ of inequalities on $\{x/\alpha\}, \dots, \{x/\alpha^n\}$ holds, and they ask (Question 1) whether the same equivalence holds for irrational $\alpha \in (1, 2.618 \dots)$. It fails on $(1, 5/4]$: if $1 < \alpha < \varphi = (1 + \sqrt{5})/2$ then every integer x with $\alpha^2/(\alpha^2 - 1) < x < 1/(\alpha - 1)$ is a fixed point of f_α , hence lies in the range of every iterate, yet $\text{RC}_{\leq 2}(\alpha, x)$ fails, and for irrational $\alpha \leq (1 + \sqrt{17})/4 = 1.2807 \dots$, in particular for every irrational $\alpha \in (1, 5/4]$, the integer $x^* = \lfloor 1/(\alpha - 1) \rfloor$ lies in that window. It does hold at level n under the sufficient condition $\alpha^n \geq \alpha^{n-1} + \dots + \alpha + 1$, equivalently $\alpha \geq a_n$ (the n -bonacci constant); since $a_n < 2$ for all n , it holds at every level for every irrational $\alpha > 2$. One direction, $\text{RC}_{\leq n} \Rightarrow$ membership, needs no hypothesis on α at all. Sharpness of a_n is not claimed: the cutoff is sufficient only, and no per-level threshold is established here.

1. QUESTION 1 AND THE ANSWER

Throughout, $\alpha > 1$ is irrational, $\lfloor t \rfloor$ and $\{t\} = t - \lfloor t \rfloor$ are the integral and fractional parts of t , and

$$f = f_\alpha : \mathbb{Z}_{\geq 1} \rightarrow \mathbb{Z}_{\geq 1}, \quad f(y) = \lfloor \alpha y \rfloor.$$

We write $\mathcal{R}_n = f^n(\mathbb{Z}_{\geq 1})$ for the range of the n -th iterate. “Natural number” means *positive* integer; this is forced by the statement below, whose first clause reads $1 - 1/\alpha < \{x/\alpha\} < 1$ and so is false at $x = 0$.

Khani, Valizadeh and Zarei [1] prove the following (their unnumbered main theorem).

Suppose that $n \geq 1$, and α is an irrational number strictly greater than $\frac{3+\sqrt{5}}{2}$. Then, a natural number x belongs to the range of f^n if and only if $1 - \frac{1}{\alpha} < \{\frac{x}{\alpha}\} < 1$ and for each $i = 2, \dots, n$ we have that

$$(1) \quad 1 - \frac{2}{\alpha} < \left\{ \frac{x}{\alpha^i} \right\} - \frac{\left\{ \frac{x}{\alpha^{i-1}} \right\}}{\alpha} < 1 - \frac{1}{\alpha}.$$

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Write $\text{RC}_{\leq n}(\alpha, x)$ for the conjunction of the first clause with (1) for $i = 2, \dots, n$, and $\text{RC}_i(\alpha, x)$ for its i -th clause. The paper [1] closes with two questions; the first is reproduced here in full.

Question 1 of [1]. Our assumption on α being strictly greater than $\frac{3+\sqrt{5}}{2} \approx 2.618$ implies that it satisfies the algebraic inequalities below:

$$\alpha^2 - 3\alpha + 1 > 0 \quad \text{and} \quad \alpha^4 - 2\alpha^3 - \alpha + 1 > 0.$$

In fact, the greatest root of the polynomials above is approximately 2.618 and 2.118, respectively. As we observed, these inequalities are crucial to prove Lemma [Technical Lemma]. A natural question is whether our theorem holds also for irrationals belonging to the following set:

$$(1, 2.618\sim) = (1, 2.118\sim) \cup (2.118\sim, 2.618\sim).$$

In this regard, recall that the function f_α is surjective whenever α is less than or equal to one.

A single counterexample answers Question 1 in the negative; the ones produced here are elementary. What is proved below says where the criterion fails, and gives a sufficient condition for it to hold.

- **No, on $(1, 5/4]$.** Theorem 7 and Corollary 8 produce, for every irrational $\alpha \in (1, 5/4]$, an integer x that lies in $\mathcal{R}_n$ for *every* n and for which $\text{RC}_{\leq n}(\alpha, x)$ is false already at index 2. The witnesses are fixed points of f_α .
- **Yes, on $(2, (3+\sqrt{5})/2)$.** Theorem 6 proves the equivalence for every irrational $\alpha > 2$ and every n , and at level n under the sufficient condition $\alpha^n \geq \alpha^{n-1} + \dots + 1$, which is not claimed to be necessary.
- **Open on $(\varphi, 2)$.** For α in that band Theorem 6 settles the levels n with $\alpha \geq a_n$ and leaves the others undecided, and we exhibit no counterexample there. At the left endpoint $\alpha = \varphi$ the equivalence itself is settled in the negative, at level 5 (Section 5); level $n = 3$ at φ is left open. See Section 6.

Nothing here contradicts the theorem of [1]: its hypothesis $\alpha > (3+\sqrt{5})/2$ is untouched, and both witnesses below lie at or below $\varphi < 2$.

The case $n = 1$ of the criterion is published as Fraenkel–Levitt–Shimshoni [5] and, in the short form used here, as Szűsz [6].

2. TWO ELEMENTARY LEMMAS

Following [1] put $v_0(x) = x$ and $v_k(x) = \lfloor v_{k-1}(x)/\alpha \rfloor + 1$, and

$$\rho_0(x) = 0, \quad \rho_k(x) = \frac{1 - \{x/\alpha^k\}}{\alpha} \quad (k \geq 1),$$

so that $\rho_k(x) \in (0, 1/\alpha)$ for $k \geq 1$, because x/α^k is irrational. Multiplying by α one checks at once that for $k \geq 1$

$$(2) \quad \text{RC}_k(\alpha, x) \iff 0 < \rho_k(x) - \frac{\rho_{k-1}(x)}{\alpha} < \frac{1}{\alpha^2},$$

the case $k = 1$ reading $0 < \rho_1(x) < 1/\alpha^2$: indeed $\alpha^2(\rho_k - \rho_{k-1}/\alpha) = \alpha - 1 - \alpha \{x/\alpha^k\} + \{x/\alpha^{k-1}\}$, and dividing the resulting double inequality $0 < \cdot < 1$ by α gives exactly (1). We use the form (2) from now on.

Lemma 1. *Let $y \geq 1$ be an integer. Then $y \in \mathcal{R}_1$ if and only if $\rho_1(y) < 1/\alpha^2$, and in that case the unique z with $f(z) = y$ is $v_1(y)$.*

Proof. $f(z) = y$ means $y \leq \alpha z < y + 1$, i.e. $y/\alpha \leq z < (y + 1)/\alpha$. As α is irrational and $y \geq 1$, $y/\alpha \notin \mathbb{Z}$, so an integer z in that window exists if and only if $\lfloor y/\alpha \rfloor + 1 < (y + 1)/\alpha$, and then it is unique and equals $v_1(y)$. Finally $\lfloor y/\alpha \rfloor + 1 < (y + 1)/\alpha$ is $1 - \{y/\alpha\} < 1/\alpha$, that is $\rho_1(y) < 1/\alpha^2$. $\square$

Lemma 2. *For all $n \geq 1$ and all integers $x \geq 1$: $x \in \mathcal{R}_n$ if and only if $v_k(x) \in \mathcal{R}_1$ for $k = 0, 1, \dots, n - 1$.*

Proof. By Lemma 1, $x \in \mathcal{R}_n$ iff $x \in \mathcal{R}_1$ and $v_1(x) \in \mathcal{R}_{n-1}$; iterate, using $v_k = v_1 \circ v_{k-1}$. $\square$

The next lemma is the whole mechanism. Say that x carries at level k if $c_k(x) := \lfloor \{x/\alpha^k\} + \rho_{k-1}(x) \rfloor$ equals 1; since $\{x/\alpha^k\} < 1$ and $\rho_{k-1}(x) < 1/\alpha < 1$, indeed $c_k(x) \in \{0, 1\}$.

Lemma 3. *Fix $x \geq 1$ and $k \geq 1$, and suppose $v_{k-1}(x) = x/\alpha^{k-1} + \alpha\rho_{k-1}(x)$. Then*

$$(3) \quad \rho_1(v_{k-1}(x)) = \rho_k(x) - \frac{\rho_{k-1}(x)}{\alpha} + \frac{c_k(x)}{\alpha},$$

$$(4) \quad v_k(x) = \lfloor x/\alpha^k \rfloor + 1 + c_k(x),$$

and $c_k(x) = 0$ if and only if $\rho_k(x) - \rho_{k-1}(x)/\alpha > 0$. Consequently, if $c_k(x) = 1$ then $\text{RC}_k(\alpha, x)$ is false, while if $c_k(x) = 0$ then

$$\text{RC}_k(\alpha, x) \iff v_{k-1}(x) \in \mathcal{R}_1 \quad \text{and} \quad v_k(x) = x/\alpha^k + \alpha\rho_k(x).$$

Proof. Dividing the hypothesis by α gives $v_{k-1}(x)/\alpha = x/\alpha^k + \rho_{k-1}(x)$, whence $\{v_{k-1}(x)/\alpha\} = \{x/\alpha^k\} + \rho_{k-1}(x) - c_k(x)$ and

$$\rho_1(v_{k-1}(x)) = \frac{1 - \{x/\alpha^k\} - \rho_{k-1}(x) + c_k(x)}{\alpha} = \rho_k(x) - \frac{\rho_{k-1}(x)}{\alpha} + \frac{c_k(x)}{\alpha},$$

because $\alpha(\rho_k(x) - \rho_{k-1}(x)/\alpha) = 1 - \{x/\alpha^k\} - \rho_{k-1}(x)$. The same identity shows that $c_k(x) = 0$, i.e. $\{x/\alpha^k\} + \rho_{k-1}(x) < 1$, is equivalent to $\rho_k(x) - \rho_{k-1}(x)/\alpha > 0$. Next,

$$v_k(x) = \lfloor \frac{v_{k-1}(x)}{\alpha} \rfloor + 1 = \frac{v_{k-1}(x)}{\alpha} + \alpha\rho_1(v_{k-1}(x)) = \frac{x}{\alpha^k} + 1 - \left\{ \frac{x}{\alpha^k} \right\} + c_k(x),$$

which is (4); when $c_k(x) = 0$ this is $x/\alpha^k + \alpha\rho_k(x)$. Finally, if $c_k(x) = 1$ then $\rho_k(x) - \rho_{k-1}(x)/\alpha \leq 0$ and RC_k fails by (2), while if $c_k(x) = 0$ then $\rho_k(x) - \rho_{k-1}(x)/\alpha = \rho_1(v_{k-1}(x))$, which is positive, so $\text{RC}_k(\alpha, x)$ reduces to $\rho_1(v_{k-1}(x)) < 1/\alpha^2$; apply Lemma 1. $\square$

3. THE POSITIVE HALF

One implication of (1) costs nothing.

Proposition 4. *For every irrational $\alpha > 1$, every $n \geq 1$ and every integer $x \geq 1$: if $\text{RC}_{\leq n}(\alpha, x)$ holds then $x \in \mathcal{R}_n$.*

Proof. Induct on $k = 1, \dots, n$ the statement “ $v_{k-1}(x) \in \mathcal{R}_1$ and $v_k(x) = x/\alpha^k + \alpha\rho_k(x)$ ”. The hypothesis of Lemma 3 holds for $k = 1$ because $v_0(x) = x = x/\alpha^0 + \alpha\rho_0(x)$, and is carried forward by the induction. At each step $\text{RC}_k(\alpha, x)$ gives $\rho_k(x) - \rho_{k-1}(x)/\alpha > 0$, hence $c_k(x) = 0$ by Lemma 3, hence $v_{k-1}(x) \in \mathcal{R}_1$ and $v_k(x) = x/\alpha^k + \alpha\rho_k(x)$. Now apply Lemma 2. $\square$

So every failure of the equivalence is of one shape: x in the range, criterion false. The converse implication is where a hypothesis on α is needed; the one used below is sufficient, and is not claimed to be necessary. For $n \geq 1$ let

$$p_n(t) = t^{n+1} - 2t^n + 1.$$

Lemma 5. *For each $n \geq 2$, p_n has exactly one root a_n in $(1, \infty)$; it satisfies $1 < a_n < 2$, and for $t > 1$*

$$t^n \geq t^{n-1} + t^{n-2} + \dots + t + 1 \iff p_n(t) \geq 0 \iff t \geq a_n.$$

Moreover $a_2 < a_3 < a_4 < \dots < 2$ and $a_n \rightarrow 2$, with

$$\begin{aligned} a_2 = \varphi = 1.6180339887\dots, \quad a_3 = 1.8392867552\dots, \\ a_4 = 1.9275619754\dots, \quad a_5 = 1.9659482366\dots, \\ a_6 = 1.9835828434\dots \end{aligned}$$

For $n = 1$ the displayed inequality reads $t \geq 1$ and holds for all $t > 1$; we set $a_1 = 1$.

Proof. Multiplying $t^n \geq \sum_{i=0}^{n-1} t^i = (t^n - 1)/(t - 1)$ by $t - 1 > 0$ gives $t^{n+1} - t^n \geq t^n - 1$, i.e. $p_n(t) \geq 0$; the steps are reversible. Now $p'_n(t) = t^{n-1}((n+1)t - 2n)$, so on $(1, \infty)$ the function p_n decreases on $(1, \frac{2n}{n+1})$ and increases on $(\frac{2n}{n+1}, \infty)$. Since $p_n(1) = 0$ we get $p_n < 0$ on $(1, \frac{2n}{n+1}]$, and since $p_n(2) = 2^{n+1} - 2^{n+1} + 1 = 1 > 0$ there is exactly one root $a_n \in (\frac{2n}{n+1}, 2)$, with $p_n \geq 0$ on $[a_n, \infty)$ and $p_n < 0$ on $(1, a_n)$. For monotonicity, $p_{n+1}(t) - p_n(t) = t^n(t-1)(t-2) < 0$ for $t \in (1, 2)$, so $p_{n+1}(a_n) < p_n(a_n) = 0$ and therefore $a_{n+1} > a_n$. For the limit, $p_n(2-\varepsilon) = 1 - \varepsilon(2-\varepsilon)^n \rightarrow -\infty$ for fixed $\varepsilon \in (0, 1)$, so $a_n > 2 - \varepsilon$ eventually. Finally $p_2(t) = t^3 - 2t^2 + 1 = (t-1)(t^2 - t - 1)$, so $a_2 = \varphi$; the four decimals are recomputed in the accompanying program. $\square$

Theorem 6. *Let $\alpha > 1$ be irrational, let $n \geq 1$, and assume*

$$(5) \quad \alpha^n \geq \alpha^{n-1} + \alpha^{n-2} + \dots + \alpha + 1, \quad \text{equivalently } \alpha \geq a_n.$$

Then for every integer $x \geq 1$,

$$x \in \mathcal{R}_n \iff \text{RC}_{\leq n}(\alpha, x).$$

In particular the equivalence holds for every $n \geq 1$ whenever $\alpha > 2$, and at level n whenever $\alpha \geq a_n$.

Proof. Proposition 4 gives $\Leftarrow$ with no hypothesis. For $\Rightarrow$ assume $x \in \mathcal{R}_n$; by Lemma 2, $v_{k-1}(x) \in \mathcal{R}_1$ for $k = 1, \dots, n$, i.e. $\rho_1(v_{k-1}(x)) < 1/\alpha^2$ by Lemma 1.

Dividing (5) by α^n turns it into $1 \geq \sum_{j=1}^n \alpha^{-j}$, i.e.

$$(6) \quad \sum_{j=2}^k \alpha^{-j} \leq \sum_{j=2}^n \alpha^{-j} \leq 1 - \frac{1}{\alpha} \quad (1 \leq k \leq n),$$

the first inequality because all terms are positive.

We prove by induction on $k = 1, \dots, n$ that $c_k(x) = 0$, that $\text{RC}_k(\alpha, x)$ holds and that $v_k(x) = x/\alpha^k + \alpha\rho_k(x)$; the last clause supplies the hypothesis of Lemma 3 at the next step, and holds for $k = 0$ trivially. For $k = 1$, $c_1(x) = \lfloor \{x/\alpha\} \rfloor = 0$, so Lemma 3 gives everything.

Let $k \geq 2$ and suppose the claim holds below k . Then $\text{RC}_j(\alpha, x)$ holds for $j \leq k-1$, so by (2) $\rho_j(x) < \alpha^{-2} + \rho_{j-1}(x)/\alpha$ for those j ; starting from $\rho_0(x) = 0$ this telescopes to

$$\rho_{k-1}(x) < \sum_{j=2}^k \alpha^{-j} \leq 1 - \frac{1}{\alpha},$$

the last step by (6). Suppose $c_k(x) = 1$. Then, by (3) and $\rho_k(x) > 0$,

$$\rho_1(v_{k-1}(x)) = \rho_k(x) - \frac{\rho_{k-1}(x)}{\alpha} + \frac{1}{\alpha} > \frac{1 - \rho_{k-1}(x)}{\alpha} > \frac{1 - (1 - \frac{1}{\alpha})}{\alpha} = \frac{1}{\alpha^2},$$

contradicting $v_{k-1}(x) \in \mathcal{R}_1$. Hence $c_k(x) = 0$, so Lemma 3 turns $v_{k-1}(x) \in \mathcal{R}_1$ into $\text{RC}_k(\alpha, x)$, and it also gives the identity $v_k(x) = x/\alpha^k + \alpha\rho_k(x)$.

If $\alpha > 2$ then $\sum_{j=1}^n \alpha^{-j} < \sum_{j \geq 1} \alpha^{-j} = 1/(\alpha - 1) < 1$, so (5) holds for every n ; alternatively $a_n < 2 < \alpha$ by Lemma 5. $\square$

4. THE NEGATIVE HALF

Theorem 7. *Let α be irrational with $1 < \alpha < \varphi$, and let x be an integer with*

$$(7) \quad \frac{\alpha^2}{\alpha^2 - 1} < x < \frac{1}{\alpha - 1}.$$

Then

- (i) $f_\alpha(x) = x$; consequently $x \in \mathcal{R}_n$ for every $n \geq 1$;
- (ii) $\lfloor x/\alpha \rfloor = x - 1$ and $\lfloor x/\alpha^2 \rfloor = x - 2$, and the quantity appearing in (1) at $i = 2$ is

$$D_2(\alpha, x) := \left\{ \frac{x}{\alpha^2} \right\} - \frac{\{x/\alpha\}}{\alpha} = \frac{x-1}{\alpha} - (x-2) = \left(1 - \frac{1}{\alpha}\right) + \left\{ \frac{x}{\alpha} \right\};$$

- (iii) $\text{RC}_1(\alpha, x)$ holds and $\text{RC}_2(\alpha, x)$ fails, on the upper side: $D_2(\alpha, x) > 1 - 1/\alpha$. Hence $\text{RC}_{\leq n}(\alpha, x)$ is false for every $n \geq 2$, while $x \in \mathcal{R}_n$.

The interval in (7) is nonempty if and only if $\alpha < \varphi$.

Proof. Nonemptiness: $\alpha^2/(\alpha^2 - 1) < 1/(\alpha - 1)$ iff $\alpha^2(\alpha - 1) < \alpha^2 - 1$ iff $\alpha^2 < \alpha + 1$ iff $\alpha < \varphi$.

(i) $x < 1/(\alpha - 1)$ gives $\alpha x < x + 1$, and $\alpha x > x$, so $\lfloor \alpha x \rfloor = x$. A fixed point of f_α lies in $\mathcal{R}_n$ for all n .

(ii) From $x < 1/(\alpha - 1) < \alpha/(\alpha - 1)$ we get $x(\alpha - 1) < \alpha$, i.e. $x/\alpha > x - 1$; also $x/\alpha < x$; so $\lfloor x/\alpha \rfloor = x - 1$. The left inequality of (7) gives $x(\alpha^2 - 1) > \alpha^2$, i.e. $x/\alpha^2 < x - 1$; and $\alpha + 1 \leq 2\alpha^2$ for $\alpha \geq 1$ gives $1/(\alpha - 1) = (\alpha + 1)/(\alpha^2 - 1) \leq 2\alpha^2/(\alpha^2 - 1)$, so $x(\alpha^2 - 1) < 2\alpha^2$, i.e. $x/\alpha^2 > x - 2$; so $\lfloor x/\alpha^2 \rfloor = x - 2$. Therefore

$$D_2(\alpha, x) = \left(\frac{x}{\alpha^2} - x + 2 \right) - \frac{1}{\alpha} \left(\frac{x}{\alpha} - x + 1 \right) = \frac{x - 1}{\alpha} - (x - 2),$$

and $(1 - \frac{1}{\alpha}) + \{x/\alpha\} = 1 - \frac{1}{\alpha} + \frac{x}{\alpha} - x + 1 = \frac{x-1}{\alpha} - (x-2)$, the same number.

(iii) RC_1 says $\{x/\alpha\} > 1 - 1/\alpha$, i.e. $x/\alpha - (x-1) > 1 - 1/\alpha$, i.e. $(x+1)/\alpha > x$, i.e. $x(\alpha - 1) < 1$: true by (7). And by (ii)

$$D_2(\alpha, x) - \left(1 - \frac{1}{\alpha} \right) = \left\{ \frac{x}{\alpha} \right\} > 0,$$

strictly, since x/α is irrational. So the right-hand inequality of (1) fails at $i = 2$. $\square$

Corollary 8. Let α be irrational with $1 < \alpha \leq \frac{1+\sqrt{17}}{4} = 1.2807764064\dots$ – in particular let $\alpha \in (1, 5/4]$ be irrational. Then $x^* = \lfloor 1/(\alpha - 1) \rfloor$ satisfies (7), so Theorem 7 applies to it: x^* is a fixed point of f_α and $\text{RC}_{\leq n}(\alpha, x^*)$ fails for every $n \geq 2$.

Proof. Put $L = 1/(\alpha - 1)$, which is irrational, so $x^* = \lfloor L \rfloor < L$: the right half of (7) holds. Also $\alpha^2/(\alpha^2 - 1) = cL$ with $c = \alpha^2/(\alpha + 1)$, and $c < 1$ because $\alpha < \varphi$. Now $x^* > cL$ is $(1 - c)L > \{L\}$, and $\{L\} < 1$, so it suffices that $(1 - c)L \geq 1$. But

$$(1 - c)L = \frac{\alpha + 1 - \alpha^2}{\alpha^2 - 1},$$

and $\alpha + 1 - \alpha^2 \geq \alpha^2 - 1$ is $2\alpha^2 - \alpha - 2 \leq 0$, i.e. $\alpha \leq \frac{1+\sqrt{17}}{4}$. Finally $5/4 < \frac{1+\sqrt{17}}{4}$. $\square$

Remark. Corollary 8 is a sufficient condition, not a characterisation: for $\alpha \in (\frac{1+\sqrt{17}}{4}, \varphi)$ the interval (7) is nonempty but need not contain an integer, and then Theorem 7 produces nothing.

A worked witness. Take

$$\alpha = \frac{7 + \sqrt{2}}{8} = 1.0517766952966\dots, \quad 64\alpha^2 - 112\alpha + 47 = 0,$$

the unique root of $64t^2 - 112t + 47$ in $(1.05, 1.06)$; it is irrational and less than $5/4$. Here $\alpha - 1 = (\sqrt{2} - 1)/8$ and $\alpha^2 - 1 = (14\sqrt{2} - 13)/64$, so (7) is

$$\frac{1055 + 896\sqrt{2}}{223} < x < 8 + 8\sqrt{2},$$

that is,

$$10.4131630129\dots < x < 19.3137084989\dots,$$

whose integers are $x = 11, 12, \dots, 19$, the largest being $x^* = \lfloor 1/(\alpha - 1) \rfloor = 19$. For each of them $f_\alpha(x) = x$, and by Theorem 7(ii)

$$D_2(\alpha, x) = \frac{9x + 38 - 8(x - 1)\sqrt{2}}{47}, \quad 1 - \frac{1}{\alpha} = \frac{8\sqrt{2} - 9}{47},$$

the latter being $0.0492278404\dots$; so $D_2(\alpha, x) > 1 - 1/\alpha$ is equivalent to $x < 9 + 8\sqrt{2} = 20.3137084989\dots$, which the nine values satisfy. An exhaustive check at this α of the equivalence $x \in \mathcal{R}_2 \Leftrightarrow \text{RC}_{\leq 2}(\alpha, x)$ over $x = 1, \dots, 24$ finds it *false* at exactly these nine values of x and *true* at the other fifteen; nothing here rules out further counterexamples at this α above $x = 24$.

5. A SECOND WITNESS: $\alpha = \varphi$, $n = 5$, $x = 22$

The witnesses of Theorem 7 all fail at level 2 and all have $\alpha < \varphi$. The equivalence also fails higher up, at α equal to the golden ratio itself, where Theorem 6 guarantees levels 1 and 2 (because $\varphi = a_2$) and permits failure from level 3 on.

Let $\alpha = \varphi = (1 + \sqrt{5})/2$, so $1/\varphi = \varphi - 1$. Then

$$3 \xrightarrow{f} 4 \xrightarrow{f} 6 \xrightarrow{f} 9 \xrightarrow{f} 14 \xrightarrow{f} 22,$$

each step being $y \mapsto \lfloor \varphi y \rfloor$, so $22 \in \mathcal{R}_5$. On the other hand, from $\varphi^{-1} = \varphi - 1$, $\varphi^{-2} = 2 - \varphi$, $\varphi^{-3} = 2\varphi - 3$, $\varphi^{-4} = 5 - 3\varphi$, $\varphi^{-5} = 5\varphi - 8$ one computes in $\mathbb{Z}[\varphi]$

$$\{22/\varphi\} = 22\varphi - 35, \quad \{22/\varphi^2\} = 36 - 22\varphi, \quad \{22/\varphi^3\} = 44\varphi - 71,$$

$$\{22/\varphi^4\} = 107 - 66\varphi, \quad \{22/\varphi^5\} = 110\varphi - 177,$$

and hence, writing $D_i = \{22/\varphi^i\} - \{22/\varphi^{i-1}\}/\varphi$,

$$D_2 = 13\varphi - 21 = 0.0344418537\dots, \quad D_3 = 8\varphi - 13 = -0.0557280900\dots,$$

$$D_4 = 5\varphi - 8 = 0.0901699437\dots, \quad D_5 = 3\varphi - 4 = 0.8541019662\dots$$

The bounds of (1) are $1 - 2/\varphi = 3 - 2\varphi = -0.2360679774\dots$ and $1 - 1/\varphi = 2 - \varphi = 0.3819660112\dots$, so D_2, D_3, D_4 satisfy (1) while

$$D_5 - \left(1 - \frac{1}{\varphi}\right) = (3\varphi - 4) - (2 - \varphi) = 4\varphi - 6 = 0.4721359549\dots > 0.$$

Thus $\text{RC}_{\leq 4}(\varphi, 22)$ holds, $\text{RC}_{\leq 5}(\varphi, 22)$ fails at $i = 5$, and $22 \in \mathcal{R}_5$: the equivalence fails at $\alpha = \varphi$, $n = 5$, $x = 22$.

Remark. The identity that supplies a closed form for f_φ^n , namely $f_\varphi^2 = f_\varphi + \text{id} - 1$, is in print: Theorem 1 of Connell [2], a lemma of Khani and Zarei [3], and axiom (A1.4) of [4].

6. WHAT IS NOT SETTLED

For α in the band $(\varphi, 2)$ we exhibit no counterexample, and Theorem 6 settles level n only when $\alpha \geq a_n$, leaving the levels n with $a_n > \alpha$ undecided in both directions. Since $\varphi = a_2$, Theorem 6 covers levels 1 and 2 at $\alpha = \varphi$ and permits failure from level 3 on; Section 5 exhibits a failure at level 5, and whether level 3 fails at φ we do not know. We do not claim that a_n is the least α for which the conclusion of Theorem 6 holds at level n : condition (5) is sufficient only, so this note establishes no per-level threshold, and the behaviour of the equivalence just below a_n is among what is left open. At $\alpha = (7 + \sqrt{2})/8$ the nine integers $11, \dots, 19$ are exactly the disagreements for $x \leq 24$ at $n = 2$, and exactly the integers of (7); nothing here rules out counterexamples above $x = 24$ that are not fixed points.

7. VERIFICATION

Every numerical assertion above is stated exactly, as an element of $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{5})$ or $\mathbb{Q}$, and each is re-derived by the program `verify.py` accompanying this note: each modulus is given by its minimal polynomial and an isolating interval, and every inequality between algebraic numbers is decided by integer arithmetic, with no floating-point comparison anywhere. The program re-checks the equivalence of (1) with the ρ -form (2); both witnesses in full, including membership of 22 in $\mathcal{R}_5$ by a preimage sieve independent of the printed chain, the closed forms, the window (7) with its nine integers, and the exhaustive table $x \leq 24$; the constants $a_2, \dots, a_6$ and the equivalent forms of (5); Theorem 7 over a family of quadratic irrationals of $(1, \varphi)$; and Theorem 6 with Proposition 4 by exhaustive checks of both directions over bounded ranges of x and n at moduli on both sides of every cutoff a_n . Those checks are bounded and a bounded check is not a proof; the theorems are carried by their proofs above. The program states in its own output what it does not cover.

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